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① Explain the difference between authentication and authorization.

→ Authentication is the process of verifying the identity of a user by checking credentials such as a username and password.

Authorization is a process of determining what resources or actions an authenticated user is allowed to access.

• Authentication occurs before authorization.

② Why should passwords never be stored in plain text?

→ Passwords should never be stored in plain text because anyone who gains access to the storage file can read and misuse them. Storing hashed passwords protect user credentials even if the database or file is compromised.

③ What is the purpose of password hashing?

→ Password hashing converts a password into a fixed length, irreversible hash value. During login, the entered password is hashed and compared with stored hash. This improves security by preventing the original password from being stored or easily recovered.

④ How does duplicate login prevention improve security?

→ Duplicate login prevention allows only one active session per ~~document~~ user account. It prevents unauthorized users from accessing the same account simultaneously, reduces account sharing and

help protect user data and application resources.

⑤ Suggest two additional security improvements for your application.

→ i) use TLS/SSL encryption to secure communication between the client and server so that data cannot be intercepted during transmission.

ii) Implement Two-Factor Authentication (2FA) by requiring a one-time verification code in addition to the password, providing an extra layer of security against unauthorized access.